

UX-0005 — Butlerly Screen and State Inventory

Version 1.0

Purpose

This document defines the complete Version 1 inventory of screens, views, overlays, dialogs, system states, and device adaptations required to implement the journeys in UX-0002, the information architecture in UX-0003, and the flows in UX-0004. It is the authoritative bridge between interaction design, wireframes, prototypes, engineering implementation, design QA, and usability testing.

1. Inventory Principles

- Every screen exists to support a defined user goal or recovery path.
- Smartphone, tablet, and desktop share one conceptual model but may use different layouts.
- A screen is not complete until empty, loading, success, error, offline, permission, and accessibility states are defined where relevant.
- Local-first behavior remains available when external services fail.
- Privacy, processing location, device trust, and synchronization status are visible when material.
- AI output must expose confidence, evidence, correction, and undo where appropriate.
- Optional synchronization, when implemented and enabled by the user, remains quiet; it is not required for core Version 1 operation.
- Destructive actions require clear consequences and recovery behavior.

2. State Taxonomy

Each screen may support these state classes:

- Initial: first-use or no-data condition.
- Default: normal populated state.
- Loading: short or extended processing.
- Partial: some content is available while other content is pending.
- Empty: no applicable records or tasks.
- Success: action completed.
- Warning: attention may be needed, but work can continue.
- Error: action failed or data may be unavailable.
- Offline: local mode is active.
- Degraded: external provider, connection, or synchronization is impaired.
- Permission Required: platform or user consent is needed.
- Locked: local device authentication is required only for a protected action or protected workspace; no Butlerly account is required for core operation.
- Read-Only: data is visible but cannot currently be changed.
- Conflict: automatic merge could not preserve meaning.
- Deleted: item is recoverable from Recently Deleted.
- Revoked: device or provider access is no longer trusted.

3. Screen Naming and IDs

Screen IDs use the format S-AREA-NNN. State variants append a suffix.

Examples:

- S-HOME-001 Home
- S-HOME-001-E Empty Home
- S-CAP-003-P Processing Receipt
- S-REC-004-C Record Conflict

4. Global Shell and Navigation

S-SHELL-001 Smartphone App Shell

Purpose: provide bottom navigation, top app bar, global Capture, Search, profile/settings access, and status entry.

Required states: default, offline, processing active, accessibility large-text adaptation.

Synchronization states apply only to a future optional sync capability.

S-SHELL-002 Tablet App Shell

Purpose: sidebar or adaptive navigation with list-detail support, global Capture and Search, and contextual toolbar.

Required states: compact width, expanded width, keyboard focus, pointer hover, offline.

S-SHELL-003 Desktop App Shell

Purpose: persistent sidebar, global toolbar, secondary navigation, multi-pane workspace, keyboard shortcuts, and window-state restoration.

Required states: narrow window, standard window, multi-window, offline. Synchronization states apply only to a future optional sync capability.

S-SHELL-004 Global Status Center

Contents: pending changes, import activity, processing jobs, synchronization delay, provider status, and safe retry actions.

Do not use as a general notification feed.

5. Onboarding and Trust Screens

S-ONB-001 Welcome

Primary action: Continue.

Secondary action: Explore Sample Data where appropriate.

States: standard, VoiceOver/screen reader, large text.

S-ONB-002 Privacy Summary

Shows local storage, optional synchronization, external AI consent, export, and deletion control.

Actions: Continue; Learn More.

S-ONB-003 Choose First Action

Choices: Capture Receipt, Import File, Explore Sample Data.

Device adaptation: smartphone emphasizes Capture; desktop emphasizes Import.

S-ONB-004 Contextual Permission Explanation

Variants: camera, photos/files, notifications, synchronization, external AI.
Must explain why the permission is needed and offer a non-blocking alternative.

S-ONB-005 First Task Completion

Shows value achieved and optional AI configuration. Device pairing and Apple synchronization are deferred beyond the core MVP and must never block local use.

6. Home Screens

S-HOME-001 Home

Modules: Continue Recent Work, Needs Attention, Recent Records, Capture, Search, noteworthy status.

States:

- S-HOME-001-E No records: offer Capture, Import, Sample Data.
- S-HOME-001-C Reserved for future optional cross-device continuity; not required for the core MVP.
- S-HOME-001-A Attention required.
- S-HOME-001-O Offline with local data available.
- S-HOME-001-D External provider degraded; future synchronization degradation is out of MVP scope.

Rules: no decorative dashboard clutter; no engagement pressure.

7. Capture and Intake Screens

S-CAP-001 Capture Options

Choices: Camera, Photo/Image, File, Manual Entry, Share Sheet Intake.

S-CAP-002 Camera Capture

Elements: live preview, receipt boundary, quality guidance, shutter, cancel, flash where supported.

States: permission denied, blur, glare, missing corners, multiple pages, low light.

S-CAP-003 Image Confirmation

Actions: Use Photo, Retake, Add Page, Crop/Rotate where supported.

S-CAP-004 Receipt Processing

Shows local or external processing location, progress, cancel behavior, preserved-source status.

States: local processing, external consent required, long-running, backgrounded, failed.

S-CAP-005 Focused Clarification

One question only. Examples: amount, date, merchant, category, transaction match.

Actions: answer, skip when safe, review later.

S-CAP-006 Saved Confirmation

Shows saved record summary, Undo, Open Record, and Review Later when applicable.

S-CAP-007 Manual Entry

Fields: amount, date, merchant/source, category, note, attachment.

Use progressive disclosure; do not mimic a full accounting form.

S-CAP-008 Share Sheet Import Preview

Shows item type, source application, preview, processing location, and destination.

States: supported, unsupported, duplicate suspected, consent required.

8. Import Screens

S-IMP-001 Import Center

Actions: Choose Files, Drag and Drop, Recent Imports.

Desktop supports bulk import; tablet moderate import; smartphone small-file intake.

S-IMP-002 Import Preview

Shows accepted files, unsupported files, expected item count, duplicate estimate, processing method, and consent requirements.

S-IMP-003 Import Mapping

For structured files only. Shows source columns, Butlerly fields, sample values, confidence, and validation.

S-IMP-004 Import Progress

Shows files processed, records created, current step, background behavior, and safe cancellation rules.

S-IMP-005 Import Summary

Counts: imported, linked, duplicates skipped, needs review, failed.

Actions: Review Exceptions, Retry Failures, Export Report, Undo Batch when allowed.

S-IMP-006 Import Failure Detail

Shows file, failure reason, preserved source, retry options, mapping options, and support diagnostics.

9. Inbox and Review Screens

S-INB-001 Inbox

Categories: Needs Review, Possible Duplicates, Missing Information, Consent Required, Import/Connection Issues, Deferred.

States: populated, empty/all caught up, offline, partial sync, high-priority security issue.

S-INB-002 Review Item

Shows problem, Butlerly recommendation, evidence, confidence, primary decision, alternatives, defer, and source inspection.

Smartphone: one item at a time.

Tablet/Desktop: list-detail.

S-INB-003 Focused Correction

Used for correcting one uncertain field or relationship.

S-INB-004 Deferred Items

Shows reason, optional reminder, date deferred, and resume action.

S-INB-005 Consent Required Item

Explains what data would leave the device, provider, purpose, alternatives, and consent options.

S-INB-006 Unresolved Sync Conflict

Only used when automatic merge cannot preserve meaning.

Shows both values, devices, timestamps, evidence, recommendation, choose/merge actions.

10. Duplicate Resolution Screens

S-DUP-001 Duplicate Comparison

Shows both records, matching fields, differing fields, source quality, links, and history.

Actions: Merge, Keep Both, Link, Replace Lower-Quality Source.

S-DUP-002 Merge Preview

Shows resulting canonical record and preserved sources before confirmation.

S-DUP-003 Merge Complete

Shows canonical record, Undo, and version history access.

11. Records and Detail Screens

S-REC-001 Records Browser

Views: All Records, Transactions, Receipts, Documents, Merchants, Accounts/Sources, Categories/Labels, Saved Views.

States: populated, empty, filtered empty, offline, loading index.

S-REC-002 Record List Item

Displays primary identity, amount/key value, date, type, status, thumbnail, and attention indicator.

S-REC-003 Record Detail

Sections:

1. Identity.
2. Butlerly Understanding.
3. Source Evidence.
4. Notes and Labels.
5. Related Records.
6. Activity and Version History.

7. Actions.

States: standard, incomplete, processing, needs review, deleted, read-only, offline, external source unavailable.

S-REC-004 Focused Edit

Supports one or more fields with source evidence nearby.

States: valid, validation warning, unsaved changes, offline queued save.

S-REC-005 Source Viewer

Displays receipt image, PDF, imported row, or original document.

Supports zoom, page navigation, accessible description, and source integrity status.

S-REC-006 Related Records

Shows receipt-transaction links, merchant, account/source, imported document, duplicate relationships, and AI explanation context.

S-REC-007 Version History

Shows understandable timeline with device, timestamp, changed fields, source, and reason.

Actions: compare, restore selected fields, restore version.

S-REC-008 Version Comparison

Shows current and prior values with changed fields clearly identified.

S-REC-009 Recently Deleted Record Detail

Shows deletion date, retention period, related-record impact, Restore, and Delete Permanently.

12. Search Screens

S-SRCH-001 Global Search

Inputs: natural language, voice where supported, structured filters, recent queries.

S-SRCH-002 Search Suggestions

Shows recent searches, suggested queries, entities, and filter completions without exposing private content unnecessarily.

S-SRCH-003 Search Results

Shows best matches, match reason, preview, record type, date/value, and attention status.

States: results, partial local-only results, no results, index updating, provider unavailable.

S-SRCH-004 Advanced Filters

Filters: date, amount, merchant, category, record type, account/source, location, status, labels.

Desktop/tablet may use persistent panel; smartphone uses sheet/full-screen filter view.

S-SRCH-005 No Results

Explains searched scope, unavailable sources, possible spelling/filter issue, and alternatives.

13. Assistant Screens

S-AI-001 Global Assistant

Supports questions about records, organization, and supported actions.

Must not replace normal navigation.

S-AI-002 Contextual Record Assistant

Preserves selected record and approved related context.

S-AI-003 AI Answer

Required structure: conclusion, evidence, confidence/uncertainty, next actions, source inspection, correction.

S-AI-004 AI Processing Consent

Shows exact data scope, provider, purpose, retention behavior, and local/manual alternative.

S-AI-005 AI Unavailable

Offers local processing, another approved provider, manual completion, or retry later while preserving input.

S-AI-006 Learned Preferences

Shows user-correction-derived preferences, reset controls, and enable/disable learning.

14. Device and Continuity Screens

S-DEV-001 My Devices

Shows trusted devices, device type, last connected, sync status, pending changes, Add Device.

States: first device, multiple devices, offline, sync delayed, revoked device visible in history.

S-DEV-002 Device Detail

Shows name, platform, trust date, last activity, key status, synchronization method, and Revoke Access.

S-DEV-003 Add Device

Methods: Apple Sync, QR Pairing, AirDrop/Encrypted Transfer.

S-DEV-004 Apple Sync Setup

Explains encrypted local copies, synchronized scope, platform account dependency, and enable action.

S-DEV-005 QR Pairing Display

Shows short-lived QR code, expiration, cancel, and device approval status.

S-DEV-006 QR Scanner

Shows scan target, permission handling, invalid/expired code, and manual alternative.

S-DEV-007 Device Approval

Shows new device name, type, fingerprint, requested data scope, Approve/Reject.

S-DEV-008 Initial Sync Progress

Shows encrypted transfer progress, local database preparation, and safe interruption behavior.

S-DEV-009 Device Ready

Shows successful trust, available data, and Continue Recent Work where relevant.

S-DEV-010 Manual Transfer Export

Creates encrypted device package for AirDrop or Files.

S-DEV-011 Manual Transfer Import

Authenticates, validates package, imports identity/data, handles expired or invalid package.

S-DEV-012 Revoke Device Confirmation

Explains consequences, requires device authentication, and confirms remaining devices are unaffected.

15. Synchronization States

S-SYNC-001 Quiet Synced State

No persistent message; status available in My Devices or Status Center.

S-SYNC-002 Pending Changes

Shows count and affected device availability without blocking local work.

S-SYNC-003 Sync Delayed

Shows last successful sync, devices that may be stale, Retry, and Manual Transfer.

S-SYNC-004 Merge Completed

Optional activity detail showing non-competing fields merged automatically.

S-SYNC-005 Same-Field Timestamp Resolution

Activity detail showing which value became current and preserving the prior value in history.

S-SYNC-006 Unresolved Conflict

Routes to Inbox only when integrity cannot be preserved automatically.

S-SYNC-007 Revoked Source Device

Changes from revoked key are rejected and recorded securely.

16. Privacy and Settings Screens

S-SET-001 Settings Home

Sections: Profile and Preferences, Privacy & Data, My Devices, Connections, AI Preferences, Notifications, Accessibility, Advanced.

S-SET-002 Profile and Preferences

Locale, currency, date format, appearance, default views.

S-PRV-001 Privacy & Data Overview

Shows local data, synchronized data, connected sources, external AI providers, trusted devices, export, and deletion.

S-PRV-002 Local Data Detail

Shows storage use, encryption state, local database status, cache management, and export.

S-PRV-003 Synchronized Data Detail

Shows scope, transport/provider, last sync, devices, disable behavior, and manual transfer.

S-PRV-004 External AI Providers

Shows approved providers, feature-level consent, data scope, last use, revoke/disable.

S-PRV-005 Connections

Shows connected financial sources, cloud drives, permissions, last access, reconnect, disconnect.

S-PRV-006 Export Setup

Scope, format, originals, version history, encryption, destination.

S-PRV-007 Export Progress

Shows local generation, authentication, cancel rules, and completion.

S-PRV-008 Export Ready

Actions: Save to Files, AirDrop, approved destination.

S-PRV-009 Delete Selected Data

Shows selected scope, relationships, recovery period, confirmation.

S-PRV-010 Erase All Data

Shows local, synchronized, backup, connected-source, and device consequences. Requires explicit confirmation and authentication.

S-SET-003 AI Preferences

Local AI availability, external provider, consent behavior, learning from corrections, reset learned preferences.

S-SET-004 Notifications

Types, urgency, privacy-safe previews, quiet hours where supported.

S-SET-005 Accessibility

Text size behavior, contrast, motion, screen-reader support, voice input, haptics where applicable.

S-SET-006 Advanced

Storage diagnostics, import logs, synchronization diagnostics, developer/troubleshooting controls.

17. Offline and Degraded Screens

S-OFF-001 Offline Status Detail

Shows local capabilities available, connected features unavailable, pending changes, and no-data-loss assurance.

S-OFF-002 Connected Feature Unavailable

Explains why a requested feature cannot run offline and offers local/manual/later options.

S-DEG-001 Provider Degraded

Shows affected feature, preserved task/input, local alternative, approved alternative provider, retry.

S-DEG-002 Connection Authentication Expired

Shows affected source, reconnect, last successful access, and local records retained.

S-DEG-003 Storage Full

Shows what is preserved, manage storage, export archive, remove cached copies, cancel.

18. Error and Recovery Screens

S-ERR-001 Generic Recoverable Error

Required content: what failed, what was preserved, whether data left device, what to do next, whether retry is safe.

S-ERR-002 Receipt Extraction Failed

Actions: Retake, Manual Entry, Another Method, Review Later.

S-ERR-003 Unsupported Format

Actions: Keep Original, Open as Document, Convert Externally, Cancel.

S-ERR-004 Import Failed

Actions: Retry, Change Mapping, Keep Original, Export Report.

S-ERR-005 Sync Failed

Actions: Retry, Inspect Pending Changes, Apple Sync Settings, Manual Transfer.

S-ERR-006 Authentication Failed

Actions: Retry Authentication, Use Another Method, Cancel.

S-ERR-007 Corrupted or Invalid Archive

Actions: Choose Another File, Verify Secret, Restore Backup, Cancel.

19. Destructive and Authentication Overlays

O-AUTH-001 Device Authentication

Used for sensitive export, device revocation, permanent deletion, erase all data, and key operations.

O-CFM-001 Standard Confirmation

Used only when consequence is meaningful and not safely reversible.

O-CFM-002 Delete Confirmation

Shows item, relationship impact, recovery window, confirm/cancel.

O-CFM-003 Permanent Delete Confirmation

Requires stronger language and authentication where appropriate.

O-CFM-004 Revoke Device Confirmation

Shows device identity and confirms other devices remain active.

O-UNDO-001 Undo Banner

Time-limited convenience; version history remains the durable recovery path.

20. Notification and External Entry States

N-001 Receipt Ready for Review

Deep-links to record or Inbox item.

N-002 Import Completed

Shows imported and exception counts without exposing private financial details on lock screen by default.

N-003 Action Required

Used for consent, connection expiration, unresolved conflict, or security concern.

N-004 Device Added or Revoked

Security notification with trusted-device context.

N-005 Continue Recent Work

Optional, non-intrusive continuity prompt.

External entry routes:

- Share Sheet → S-CAP-008.
- File Open With Butlerly → S-IMP-002.
- Pairing QR → S-DEV-006.
- Exception notification → S-INB-002.
- Search shortcut → S-SRCH-001.

21. Responsive and Native Adaptation Matrix

Smartphone

- Bottom navigation and prominent Capture.
- Single-column task flows.
- One-at-a-time review.
- Full-screen camera, filters, and focused edits.
- Sheets for short choices and consent.

Tablet

- Sidebar and list-detail layouts.
- Side-by-side comparison and source viewing.
- Touch, keyboard, pointer, drag and drop.
- Adaptive compact/expanded widths.

Desktop

- Persistent sidebar, toolbar, secondary navigation.
- Multi-pane lists, details, comparisons, and settings.
- Keyboard-first operation and shortcuts.
- Bulk import, multi-select, export, diagnostics.
- Window and panel-state restoration.

A different layout does not create a different screen ID unless the user goal, information hierarchy, or behavior materially changes.

22. Accessibility State Requirements

Every applicable screen must define:

- Logical focus order.
- Accessible names and roles.
- Dynamic status announcements.
- Keyboard completion path.
- Touch target compliance.
- Non-color status indicators.
- Large-text reflow without clipping.
- Reduced-motion behavior.
- Screen-reader descriptions for receipt images, comparisons, confidence, and changed fields.
- Focus return after dialogs and overlays.
- Durable recovery when a time-limited Undo expires.

23. Screen-to-Flow Traceability

F01: S-ONB-001 through S-ONB-005.
F02: S-CAP-001 through S-CAP-006.
F03: S-CAP-008, S-CAP-004 through S-CAP-006.
F04: S-IMP-001 through S-IMP-006.
F05: S-CAP-004, S-AI-003, S-INB-002.
F06: S-INB-001 through S-INB-004.
F07: S-DUP-001 through S-DUP-003.
F08: S-SRCH-001 through S-SRCH-005, S-REC-003.
F09: S-AI-002 through S-AI-005.
F10: S-REC-004, S-REC-007, S-REC-008, O-UNDO-001.
F11: S-DEV-001 through S-DEV-011.
F12: S-SYNC-001 through S-SYNC-007.
F13: S-DEV-002, S-DEV-012, O-AUTH-001.
F14: S-OFF-001, S-OFF-002, global offline variants.
F15: S-PRV-001 through S-PRV-005, S-AI-004.
F16: S-PRV-006 through S-PRV-008.
F17: S-PRV-009, S-PRV-010, S-REC-009, delete overlays.
F18: S-ERR-001 through S-ERR-007 and degraded states.

24. Version 1 Priority

P0 — Required for MVP usability and trust

- App shells.
- Onboarding and privacy summary.
- Home.
- Receipt capture and processing.
- Import preview/progress/summary.
- Inbox and review item.
- Records browser and detail.
- Search and results.
- Contextual Assistant and consent.
- Local privacy, export, deletion, and protected-action device authentication.
- Offline, error, undo, export, delete, and local device-authentication states for sensitive actions.

P1 — Required before broad release

- Duplicate comparison and merge preview.
- Version comparison and selective restore.
- Optional backup/restore and portability refinements.
- Advanced filters.
- Connection management.
- Accessibility settings and diagnostics.

P2 — May follow initial release

- Rich saved views.
- Extensive bulk actions.
- Multi-window desktop optimization.
- Advanced AI learned-preference management.
- Cross-device synchronization, device trust, pairing, and synchronization activity history.

25. Design and Engineering Acceptance Checklist

A screen is implementation-ready when:

- It has a unique ID and purpose.
- Entry and exit paths are defined.
- Primary and secondary actions are defined.
- Normal, empty, loading, offline, error, and success states are covered where relevant.
- Permission and consent behavior is defined.
- Destructive consequences and recovery are defined.
- Smartphone, tablet, and desktop behavior is specified.
- Accessibility requirements are testable.
- Analytics events exclude private content by default.
- Flow and journey references are traceable.
- Local-first and external-provider failure behavior is clear.
- AI confidence, evidence, correction, and undo are present where required.

26. Dependencies and Governance

UX-0005 depends on:

- UX-0001 — Butlerly Experience Blueprint.
- UX-0002 — Butlerly Version 1 User Journey Map.
- UX-0003 — Butlerly Information Architecture and Navigation Model.
- UX-0004 — Butlerly Core User Flows.
- Product MVP scope and personas.
- Architecture requirements for local storage, encryption, synchronization, identity, and AI providers.

UX-0005 governs:

- UX-0006 — AI Interaction and Explainability Specification.
- UX-0007 — Design System Foundations.
- UX-0008 — Cross-Device Experience and Continuity.
- UX-0009 — Accessibility Standard.
- UX-0010 — Low-Fidelity Wireframe Specification.
- Prototype coverage, engineering UI stories, and design QA test cases.

27. Acceptance Criteria

This inventory is complete when:

- Every P0 Version 1 flow maps to defined screens and states.
- All primary destinations and global capabilities are represented.
- Smartphone, tablet, and desktop adaptations are explicit.

- Offline, degraded, permission, conflict, deleted, and recovery states are included.
- Core MVP completion does not require account, device-trust, pairing, or synchronization screens. Cross-device synchronization and related conflict-resolution UX are deferred and may be maintained as future-design references only.
- Designers can create wireframes without inventing missing screens.
- Engineers can derive UI routes, components, state machines, and acceptance tests.
- QA can test normal, failure, recovery, privacy, and accessibility behavior.

BDS-0000 Conformance

BDS-0000 — Butler Design System v1.0 governs visual tokens, concrete theme values, typography, spacing/radius, responsive breakpoints, component/state contracts, accessibility implementation, localization/original-data presentation, and Flutter UI implementation. Existing S-INB-* and S-REC-* identifiers may remain for traceability, but visible Finance V1 primary navigation uses Home, Transactions, Review, Search, and Settings; Assistant is optional/non-primary. Device-trust and synchronization screens in this inventory are future/optional capability references and are not required for the core MVP unless product scope explicitly enables them. Core operation requires no Butlerly account. UI-authored text follows the selected application language; user-entered/imported/scanned/source data preserves its original language by default, and original currency/amount remains authoritative with any reference conversion shown separately.